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Implants and methods for treating Charcot foot and other conditions

Abstract

A fixation system for immobilizing a skeletal structure is provided. It includes an internal fixation system having a rod-plate subsystem, a shaft subsystem, and a midfoot plate subsystem. It further includes an external fixation system adapted to connect to the internal fixation system and having a sole that provides a weight-bearing platform underneath a patient's foot to enable a patient to walk with the fixation system.

Inventors:	Early; John (Dallas, TX), Pomeroy; Gregory (Gorham, ME)
Applicant:	BESPA GLOBAL, LLC (Cedar Grove, NJ)
Family ID:	1000008782962
Assignee:	BESPA GLOBAL, LLC (Cedar Grove, NJ)
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Primary Examiner: Robert; Eduardo C

Assistant Examiner: Green; Michelle C

Attorney, Agent or Firm: Corner Counsel, LLC

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of, and claims priority to, U.S. Nonprovisional patent application Ser. No. 16/963,775 filed on Jul. 21, 2020, which is a national stage of, and claims priority to, International Patent Application No. PCT/US2019/016257 filed on Feb. 1, 2019, which claims the benefit of U.S. Provisional Application Ser. No. 62/626,324, filed Feb. 5, 2018, the disclosure of which is expressly incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) The present invention relates to implants and methods for the treatment of various medical conditions typically involving mammalian extremities. One such condition is a human extremity deformity known as Charcot foot.

BACKGROUND

(2) The terminology and descriptions contained herein are principally within the art field of, and for those skilled in the art of, orthopedic medicine. As such, only brief explanations of known subject matter within this art field will be provided because the details will be well known to those skilled in this art. The present invention, however, will be thoroughly described.

(3) Damage or dysfunction of peripheral nerves of the foot causing numbness or weakness, also known as neuropathy, can lead to a condition known as Charcot arthropathy, or more commonly referred to as Charcot foot. More specifically, when a patient with neuropathy has an injury to their foot, the neuropathy may prevent them from sensing the injury. Without this defense mechanism which would otherwise cause the patient to feel pain, avoid continued injury and/or seek medical attention, the patient will continue to walk on the injured foot. This typically exacerbates the injury and affects surrounding areas of the foot, ultimately leading to possible deformity, disability, and even amputation of the foot in extreme cases.

(4) One common symptom of advanced Charcot foot is the collapse of certain joints in the foot and a resulting disfigurement of the foot. Surgical treatment often involves the re-alignment and fixation of various bones within the foot to correct such deformity.

(5) Various general internal fixation systems involving screws, plates, bolts, nails, and the like, are known and available for use to correct Charcot foot deformities. Similarly, various general external fixation systems involving external frames, pins, wires, screws, and the like, are also known and available for use to correct Charcot foot deformities. Some of the challenges in the art are constructing a customized patient solution to Charcot foot that includes internal fixation interacting with external fixation, and providing a weight-bearing platform for the affected limb, enabling a patient to walk soon after surgical intervention. The present invention offers solutions to these challenges and contemplates various novel and non-obvious combinations of implant modularity, interaction between internal and external fixation systems, and a weight-bearing platform.

SUMMARY OF THE INVENTION

(6) A fixation system is provided for immobilizing a skeletal structure, the fixation system having

an internal fixation system with one or more of a rod-plate system and a shaft system, and an external fixation system. The rod-plate system includes a rod affixed to a plate, the rod being adapted to be positioned in a bone canal, and the plate being adapted to be positioned on bone near the bone canal. The shaft system includes a shaft with a longitudinal axis, a slot on the shaft oriented in the direction of the longitudinal axis, and a hole on the shaft oriented at an angle to the longitudinal axis, the shaft further adapted to be positioned in a bone canal and configured to move two bone segments that comprise the bone canal toward each other. The external fixation system comprises a frame connected to a sole, the sole having a bottom adapted to contact ground.

(7) The external fixation system further includes a pin to connect the rod-plate system or the shaft system to the external fixation system when the rod-plate system or the shaft system is located in bone. Optionally, the rod-plate system may be connected to the shaft system when both systems are located in bone. The rod of the rod-plate system can be modularly comprised of multiple segments that are joinable by a connection. This connection can be a threaded connection or a Morse-taper connection. The plate of the rod-plate system includes a first side adapted to face bone, an opposite second side, a length, a width, a plate axis along the length, and a projection extending from the first side. This projection has an opening to communicate with the rod of the rod-plate system. This opening communicates with the rod via a threaded connection. The opening is cylindrical and has a longitudinal opening axis, where the opening is oriented such that the opening axis is at an angle to the plate axis. Finally, the shaft of the shaft system can be modularly comprised of multiple segments each joinable at a connection. The fixation system can also include a midfoot plate system attached to bone, where the midfoot plate system comprises a plate and fasteners to attach the plate to bone.

(8) Another embodiment of the fixation system that is provided for immobilizing a skeletal structure comprises an internal fixation system having a rod-plate system, a shaft system and an external fixation system. The rod-plate system includes a rod affixed to a plate, the rod being adapted to be positioned in a bone canal, and the plate being adapted to be positioned on bone near the bone canal, where the plate further includes a first side adapted to face bone, an opposite second side, a length, a width, a plate axis along the length, and a projection extending from the first side. This projection has a cylindrical opening with a longitudinal opening axis, where the opening is oriented such that the opening axis is at an angle to the plate axis. The rod-plate system is connected to the shaft system with at least one fixation element when both systems are located in bone, and the external fixation system may also be connected to one of the rod-plate system or shaft system.

(9) Another embodiment of a fixation system is provided for immobilizing a skeletal structure, this fixation system having an internal fixation system including one or more of a rod-plate system and a shaft system, and an external fixation system.

(10) The rod-plate system includes a rod affixed to a plate, the rod being adapted to be positioned in a bone canal, and the plate being adapted to be positioned on bone near the bone canal. The shaft system includes a shaft with a longitudinal axis, a slot on the shaft oriented in the direction of the longitudinal axis, and a hole on the shaft oriented at an angle to the longitudinal axis. The shaft is further adapted to be positioned in a bone canal and configured to move at least two bone segments through which the bone canal passes, toward each other. The external fixation system includes a frame connected to a sole, the sole having a housing and a bottom adapted to contact ground, where the housing contains a liner. The liner also includes an inflatable bladder in a shell. The sole can be connected to the frame with adjustable struts. The rod of the rod-plate system is modularly comprised of multiple segments each joinable by a connection, where the connection is either a threaded connection, a Morse-taper connection, or other type of connection.

(11) The external fixation system further includes at least one pin to connect the external fixation system to either the rod-plate system or shaft system, and a frame connected to a sole, where the sole has a bottom adapted to contact ground. The fixation system can also include a midfoot plate

system attached to bone, the midfoot plate system comprising a plate and a fastener.
(12) Other features of the present invention will become more apparent after a review of the Detailed Description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a fully assembled fixation system of a preferred embodiment of the present invention incorporating internal fixation, external fixation, and a weight bearing platform;
- (2) FIG. 1A is a perspective view of a human foot;
- (3) FIG. 2 is a side elevational view of a rod component of the internal fixation system shown in FIG. 1;
- (4) FIG. 3 is a side elevational view of an intrabody connection mechanism facilitating modularity of the rod component of the internal fixation system;
- (5) FIG. 4 is a side elevational view of an alternative connection mechanism to that shown in FIG. 3;
- (6) FIG. 5 is a side elevational view of an alternative end of the rod component;
- (7) FIG. 6 is a top view of a connection plate component of the internal fixation system shown in FIG. 1;
- (8) FIG. 7 is a side elevational view of the connection plate component;
- (9) FIG. 8 is a top view of an alternate embodiment of the connection plate component;
- (10) FIG. 9 is a side elevational view of the connection plate component shown in FIG. 8;
- (11) FIG. 10 is a front exploded view of the connection plate component shown in FIG. 8 with several fasteners to be used to attach the plate component to bone;
- (12) FIG. 11 is a cross-sectional view of the connection plate component shown in FIG. 8 taken along section line 11-11, connected with the rod component of FIG. 2, which is also shown in cross-section for consistency;
- (13) FIG. 12 is a cross-sectional view similar to that of FIG. 11, but depicting an alternative connection of the rod component to the plate component;
- (14) FIG. 13 is a top view depicting a human foot with several assembled rod-plate constructs of the internal fixation system;
- (15) FIG. 14 is a side elevational view of what is shown in FIG. 13;
- (16) FIG. 15 is an exploded elevational view of a shaft component of the internal fixation system;
- (17) FIG. 16 is a partial exploded elevational view of the shaft component of FIG. 15 shown assembled, along with several fasteners used to attach the shaft component to bone;
- (18) FIG. 17 is a side elevational view of a fully assembled shaft component affixed to two bones prior to reduction;
- (19) FIG. 17A is a side elevational view of what is shown in FIG. 17, depicting motions that reduce the space between the two bones;
- (20) FIG. 18 is a top view of two shafts implanted in a foot;
- (21) FIG. 19 is a side elevational view of what is shown in FIG. 18;
- (22) FIG. 20 is a top view of various internal fixation systems implanted in a foot, including plate-rod constructs, and rods connected to shafts;
- (23) FIG. 21 is a side elevational view of what is shown in FIG. 20;
- (24) FIG. 22 is a perspective view of a midfoot plate component of the internal fixation system;
- (25) FIG. 23 is a top plan view of the midfoot plate component implanted in a foot along with the other internal fixation systems depicted in FIGS. 20 and 21;
- (26) FIG. 24 is a top view of two shaft systems of the internal fixation system implanted in a foot

and connected to an external fixation system;

(27) FIG. 25 is a perspective view of a sole component that is part of the external fixation system; and

(28) FIG. 26 is an exploded perspective view of various components of the external fixation system, including the sole component forming the weight bearing platform.

DETAILED DESCRIPTION

(29) For convenience and efficiency of explanation only, the following descriptions of various embodiments of the present invention will be provided with reference to a human foot. However, this part of the body is only meant to be exemplary, non-limiting, and facilitative of a straightforward explanation of the invention, since aspects of the present invention are also envisioned to apply to other skeletal structures.

(30) Referring to FIG. 1, a fixation system 10 for treating Charcot foot and other extremity deformities is depicted. Fixation system 10 comprises an internal fixation system 40 and an external fixation system 80. Internal fixation system 40 is comprised of three general subsystems referred to as rod-plate system 50, shaft system 60, and midfoot plate system 70. External fixation system 80 relevantly includes an optional weight-bearing platform, referred to as sole 400. The various systems, and certain of their various internal and external components, are modular, and may be mixed, matched and used together or independently, as will be discussed in more detail below.

(31) With reference to FIG. 1A, the human foot comprises the forefoot 12 which includes the metatarsals 14 and phalanges 16, the midfoot 18 which includes the three cuneiform bones 20, the cuboid 22 and the navicular bone 24, and the hindfoot 26 which includes the talus 28 and calcaneus 30. For the sake of conciseness, and because anatomy is well known to those skilled in the art, further reference to the various parts of the foot may be made herein without accompanying reference numbers.

(32) Directional and spatial anatomical terminology that is also used herein is similarly well known to those skilled in the art. For instance, the term “medial” typically means closer to the midline of the body, and “lateral” typically means farther from the midline of the body. Further terms, such as “proximal”, “distal”, “anterior”, “posterior”, “superior”, “inferior”, and other such terms shall have their common and ordinary meanings in the art.

(33) As used herein, the terms “rod” and “shaft” are chosen to describe the longitudinal members of the internal fixation system for convenience and efficiency of explanation only, and are not meant to be limiting. Thus, “rod” and “shaft” are intended to be non-limiting generic terms that may include such things as a bolt, nail, screw, strut, beam and the like.

(34) As an overview, FIGS. 2-14 depict components of rod-plate system 50, which is one of three subsystems of internal fixation system 40. FIGS. 15-19 depict components of shaft system 60, which is the second of the three subsystems of internal fixation system 40. FIGS. 22-23 depict midfoot plate system 70, which is the third subsystem of internal fixation system 40. FIGS. 20, 21, and 23 depict various combinations of the various internal fixation subsystems. And FIGS. 24-26 depict components of the external fixation system 80.

(35) With reference to FIGS. 2-5, rod 100 includes a body 108 with a first end 104 and a second end 106, which terminates at a flat surface 134. As alternatively depicted in FIG. 5, second end 106 may terminate at a rounded surface 136. As will be discussed later, rounded surface 136 is better suited to facilitating the insertion of rod 100 into a bone canal.

(36) Rod 100 may be unitary and continuous from end 104 to end 106, or comprised of two or more joined segments, such as first segment 110 joined at connection 102 to second segment 112. Connection 102 generally represents various connection mechanisms that enable the joining of two rod segments together, and will be discussed in more detail, below.

(37) Body 108 of rod 100 may have a generally circular transverse cross-sectional shape or may have any other cross-sectional shape such as oval, polygonal or otherwise, as may be suited for

various applications. Body **108** may also be roughened, knurled, or otherwise provided with any other surface topography known in the art, for various purposes also known in the art, such as to provide an improved surface for bone adhesion. Furthermore, rod **100**, or any of its segments, may be solid or hollow.

(38) The first and second segments **110**, **112** of rod **100** are joined together at connection **102**. FIG. **3** depicts a Morse taper-type connection, and FIG. **4** depicts a threaded connection, both of which will be described in more detail, below. Of course, other alternative connections are envisioned. It is further envisioned that in the case of multiple segments of a rod, the same type of connection, or different types of connections, may be employed to join each two segments of a rod **100** together, depending on a wide array of factors, as known to those skilled in the art. Similarly, if it is desired to use more than one rod **100** in an implantation, it is envisioned that one segmented rod may be assembled with one type of connection and another segmented rod may be assembled with another type of connection. The rationale for doing so may be due to, for example, the various details and advantages of one type of connection over another, and the resultant properties of the assembled rods with the different connection types.

(39) As mentioned above, FIG. **3** depicts a Morse-taper type of connection. Second rod segment **112** has a male end **118**. Male end **118** includes a conical shoulder **120** that tapers to a projection **122** having a smaller diameter than body **108**. First rod segment **110** has a female end **114** comprising a conical opening **115** significantly decreasing in diameter and leading to an inner bore **116**. In general, female end **114** is dimensioned to matingly receive male end **118**. Thus, shoulder **120** has a general correspondence and sizing relationship to opening **115**, and similarly, projection **122** to bore **116**. More specifically, as is known in the art of Morse tapers, one or both of male end **118**'s shoulder **120** and projection **122** have circumferential dimensions that are slightly larger than those of the corresponding female end **114**'s opening **115** and bore **116**. When male end **118** is inserted into female end **114**, the interference in dimensions results in a press-fit, locking second rod segment **112** to first rod segment **110**.

(40) If it is desired to not have a rotational preference to the axial alignment, or no keying effect, of first rod segment **110** to second rod segment **112**, then the Morse-taper mating surfaces described just above would all have circular cross sections. Of course, if a keying effect would be desired, then the cross-sectional shapes of the mating surfaces may be oval, polygonal, or any other shape known in the art.

(41) FIG. **4**, as an alternative embodiment, depicts a threaded connection mechanism **102** between first rod segment **110** and second rod segment **112**. Female end **114** of first rod segment **110** comprises a bore **123** with internal threads **124**. Male end **118** of second rod segment **112** comprises a corresponding projection **126** with external threads **128**. Projection **126** with external threads **128** is configured to be matingly received in a threaded fashion in threaded bore **123** of female end **114**. When so assembled, first rod segment **110** is securely connected to second rod segment **112**.

(42) With reference to FIG. **5**, rod **100** may have one or more through-holes **130** configured to receive a fastener (not shown), such as a bone screw, or other fastener known in the art. As will be evident to those skilled in the art, through-hole **130** may be internally threaded so as to threadably receive a corresponding threaded fastener, or may be provided with any other connection mechanism known in the art. In other embodiments, through-hole **130** is smooth and does not contain any form of connection mechanism. Through-hole **130** can receive a fastener to facilitate fixation of rod **100** within the medullary canal within which rod **100** may be positioned.

Alternatively, through-hole **130** may receive a fastener that is also connected to one or more other rods **100** which themselves may be positioned and fixed relative to other human bones in the foot to provide fixation across such bones. In yet other embodiments, hole **130** may receive fixation elements from external fixation system **80**.

(43) Slot **132** is depicted on second end **106** of rod **100**, and its length is oriented generally along

the long axis of rod **100**. The purpose and function of slot **132** will be discussed in more detail with reference to FIGS. **15-17A**, where shaft **200** has a relatively similar slot **240**.

(44) With reference to FIG. **2**, first end **104** of rod **100** includes a projection **138** having external threads **140**, and a shoulder **142**. As will be described in more detail below, projection **138**, external threads **140**, and shoulder **142** communicate with connection plate **150** (FIG. **11**) to form an improved fixation construct. In other embodiments, connection mechanisms other than threads are envisioned such as, for instance, interference fit, cross-screw (FIG. **12**), nut-and-bolt, ratchet mechanism, and other connection methods known in the art.

(45) In its contemplated embodiments, rod **100** may be formed of any suitable material known in the art, such as titanium, or other biocompatible materials having mechanical properties suitable for the contemplated uses of fixation system **10**. Furthermore, rod **100** may be coated with any suitable biocompatible coating known in the art, such as hydroxyapatite or the like, or may be uncoated, as needed to suit particular mechanical and clinical needs.

(46) In the contemplated embodiments of the present invention, rods **100** may be provided in various lengths and configured to provide axial stability to the bones in which they reside. In some embodiments, a rod **100** is configured to have a length such that the rod **100** extends from a portion of a metatarsal into the talus or calcaneus of the hindfoot to provide axial stability. Rods **100** may also be provided in various diameters such as, for instance, a diameter of about 3 mm, about 3.5 mm, about 4 mm, about 4.5 mm or any other diameter known in the art that is suitable for the intended use of rod **100**. As will be evident to one skilled in the art, the selection of one or more appropriately sized rods **100** may be made by the surgeon in the operating room during surgery, or the rods **100** may be selected and prepared prior to surgery.

(47) FIGS. **6-12** depict embodiments of a connection plate component **150** that are configured to facilitate connection between a rod, such as rod **100**, and a bone, to improve the robustness and rigidity of an internal fixation construct of fixation system **10**. As described in an exemplary embodiment herein, plate **150** attaches to a metatarsal of a human foot. However, the description of plate **150** in the exemplary embodiment is presented for convenience and efficiency of explanation only, and is not meant to be taken as limiting the scope of the invention to the exemplary embodiment. In other embodiments, plate **150** may be fixed to bones other than the metatarsal of a human foot. For instance, plate **150** may be fixed to a metacarpal bone of a human hand or to another human bone. Plate **150** may be generally parallelogram, square, rectangular, oval, or any other shape known in the art that is suitable for fixation to the bone to which plate **150** is to be affixed.

(48) Referring to FIGS. **6-12**, plate **150** includes a first side **152**, a second side **154** opposite first side **152**, a thickness **T** extending therebetween, and a perimeter edge **156**. In an embodiment, plate **150** includes four lobes **158** each having a lobe edge **160** adjacent lobe **158** and an aperture **162** that extends through the thickness **T** of the plate **150**. In other embodiments, plate **150** may have more or fewer lobes **158**, while in still other embodiments, plate **150** may have four lobes, wherein lobes **158** includes more or fewer apertures **162**. Apertures **162** are configured to receive fasteners, such as screws **164**, and are generally circular in shape, however other shapes for one or more apertures **162** may be used, such as slots, for example. Apertures **162** include a first flared portion **166** on first side **152** of plate **150** and a second flared portion **168** on second side **154** of plate **150**. As will be understood by those skilled in the art, second flared portions **168** are configured to receive a fastener head therein so that the fastener head may be partially or fully recessed within plate **150**. Additionally, the combination of flared portions **166**, **168** together, enables for angular placement of a fixation element, such as screw **164**, through plate **150**. In various embodiments, the flared portions **166**, **168** include a gradual taper or conical shape, a recessed or step-down area, or a curved or rounded portion, or any other flare known in the art to receive the head of a fastener. In other embodiments, the flared portions **166**, **168** may be omitted so that apertures **162** extend perpendicular to thickness **T**. In such embodiments, fasteners without protruding heads can be used.

Additionally, it is envisioned that the one or more apertures **162** may be designed to accommodate a snap ring (not shown) or other mechanism to prevent a screw **164** from backing out of plate **150**. Such back-out mechanisms are well known in the art.

(49) Referring to FIG. **6**, embodiments of plate **150** may be of various shapes and configurations. Plate **150** includes a long axis A, a short axis C, a length L1 measured as the largest length along the long axis A, and a width W1 measured as the largest width along the short axis C. The shape of the exemplary plate **150** is generally a parallelogram, however the shape and configuration of plate **150** can be varied. For instance, reducing the length L1 or increasing the width W1 would result in a parallelogram shape that is shaped more like a square. In other embodiments, length L1 can be increased and width W1 can be reduced resulting in a rectangularly shaped parallelogram having one pair of opposing sides of perimeter edge **156** longer than the other pair.

(50) The length L2 is the shortest distance between opposing sections of perimeter wall **156** at the intersection of the long axis A and the perimeter wall **156**, and width W2 is the shortest distance between opposing perimeter walls **156** at the intersection of the short axis C and the perimeter wall **156**. As depicted in FIG. **6**, in an embodiment, the length L2 and width W2 cooperate to provide a generally parallelogram shape. In some embodiments, the length L2 can be increased or reduced to provide a shape that is more rectangular or more square, respectively. In other embodiments, the width W2 can be increased or reduced to provide a shape that is more square or more rectangular, respectively.

(51) The location of the lobes **158** can also be varied. Length L3 is the length between the center of two apertures **162** which are both located on the same side of the short axis C, and width W3 is the width between the same apertures **162**. In some embodiments, length L3 can be increased to create a shape that is more diamond-like. In other embodiments, length L3 can be reduced, which results in a shape that is less like a parallelogram. In some embodiments, length L3 can be reduced to zero such that the two apertures **162** are axially aligned along the short axis C, that is, both apertures **162** are the same distance from the short axis C. In some embodiments, the width W3 can be increased to create a shape that is more like a square. In other embodiments, width W3 can be reduced, which results in a shape that is more like a rectangle.

(52) The location of each aperture **162** within its lobe **158** relative to the lobe edge **160** can also be modified. In some embodiments, each aperture **162** can be disposed closer to its respective lobe edge **160**, while in other embodiments, each aperture **162** can be disposed farther from its respective lobe edge **160**. In still other embodiments, the positions of the apertures relative to their respective lobe edges **160** may be mixed and matched, as may be desirable based on a variety of factors, such as for example, if plate **150** is made as a custom patient-matched implant that necessitates an altered configuration.

(53) With reference to FIG. **6**, S1 is the shape of the periphery of plate **150** along the C axis. S2 is the shape of the periphery of plate **150** along the A axis. These shapes S1, S2, are depicted as concavities on the periphery of plate **150**. S1, S2 may be complexly formed splines created from transitions of radii of different lengths, or shaped in any other way known in the art, including having no curvature and being straight. In some embodiments, the concavities of S1 and S2 may be grossly exaggerated toward the center of the plate **150**, thus resulting in there being less plate **150** material. In other embodiments, one or both of shapes S1 and S2 may be convexities.

(54) As will be evident to those skilled in the art, there are a myriad of possible configurations of plate **150**. This further facilitates the customizability of plate **150**. For instance, plate **150** may be provided with greater or lesser L3 and W3 dimensions so as to provide more or less distance, respectively, between the apertures **162**. Similarly, others of the foregoing dimensions may be arranged to provide an altered plate **150** according to the surgeon's preference or the particular patient's needs. In some embodiments, various dimensions may be modified together to alter various aspects of plate **150**, such as the locations of lobes **158** relative to perimeter edge **156**.

(55) For the foregoing description, it should be understood that, while various lengths and widths

were depicted on only one side of axis A or C in FIG. 6, such depiction was made for clarity and efficiency of explanation only, and is not meant to be limiting. As such, for any particular length or width described, the same aspects and dimensions on the opposite side of axis A or C exist and can be modified as described above. It should further be understood that the foregoing dimensions do not need to be symmetrical and can be varied from one side of axis A or C relative to the other side of axis A or C, respectively.

(56) Continuing to refer to FIGS. 6-12, plate **150** may include a projection **180** that extends generally orthogonally away from first side **152** of plate **150**. The purpose of projection **180** is to enable plate **150** to be connected to a rod **100**. Projection **180** includes an inner opening **182** which may be configured to receive a first end **104** or second end **106** of the rod **100** of FIGS. 2-4, and also includes an external face **184** (FIG. 10). In an embodiment, inner opening **182** may be formed with internal threads **186** to communicate with external threads **140** of projection **138** of rod **100** (see FIGS. 2 and 11). In some embodiments, inner opening **182** is a blind bore (see, for instance, FIGS. 7 and 9), while in other embodiments, inner opening **182** may extend completely through projection **180**. In still other embodiments, inner opening **182** may be unthreaded, and the connection between opening **182** and rod **100** may be a Morse-taper type connection.

(57) With reference to FIGS. 2 and 11, as will be evident to those skilled in the art, during assembly of rod **100** to plate **150**, as rod **100** is inserted into opening **182** to a sufficient depth, the shoulder **142** of rod **100** will contact face **184** of projection **180** of plate **150**, thereby halting the movement of rod **100** into projection **180** and ensuring that rod **100** does not pass beyond projection **180**.

(58) In still other embodiments, plate **150** and, optionally, rod **100** may be configured with other structures or mechanisms known in the art that facilitate connection of plate **150** to a rod **100**.

(59) Referring to the exemplary embodiment of FIG. 12, rod **100** does not have threads **140** on its projection **138** but has a through-hole **178** passing transversely through projection **138**. Plate **150** is provided with a partially threaded transverse hole **170** having threads proximal to second side **154** of plate **150**. Transverse hole **170** is configured to receive a partially threaded screw **176** and threadably communicate with screw **176** in the region shown as **172**. After rod **100** is inserted into opening **182** of projection **180**, rod **100**'s through-hole **178** is axially aligned with transverse hole **170**. Optionally, this alignment may be further facilitated by having cooperating keyed surfaces on projection **138** and opening **182**, as is known in the art. Then, screw **176** is inserted into hole **170** and through rod **100**'s through-hole **178**, and threadably fixed to plate **150** at region **172**. A portion of hole **170** may be countersunk, or otherwise formed, to accept in whole or in part, a head of a fastener, such as head **174** of screw **176**, to enable head **174** to be countersunk into the projection **180** of plate **150**.

(60) Referring to FIGS. 6-8 and 11, opening **182** in projection **180** of plate **150** is formed around longitudinal axis D (FIG. 7). In this embodiment, axis D is in the same plane as axis A, which plane is defined by the cross-section line **11-11**. Axis D travels through point P which is a point in space within opening **182**. Line E is a reference line drawn parallel to axis A of plate **150**, and offset from axis A by virtue of also traveling through point P. As better seen in FIG. 11 where rod **100** is assembled with plate **150**, angle α , which is the angle between axis D and line E, represents the angle between rod **100** and plate **150**. Angle α may range anywhere from about 5° to about 30° with reference to line E, or may be any other angle that may suit various clinical needs, as is evident to those skilled in the art. In an alternate embodiment (not shown), angle α may even be a negative angle, and in such case, plate **150** would be shaped to accommodate rod **100** projecting through and above its second side **154**. For example, plate **150** may have an opening through its thickness T to accommodate rod **100**, or perimeter edge **156** of plate **150** may be concavely shaped so as to make room for rod **100** to pass adjacent to plate **150**.

(61) In other alternative embodiments, the orientation of rod **100** to plate **150** may be different. For example, axis D may be oriented at an angle to the plane formed by cross-section line **11-11**. This is depicted in FIG. 6 with reference to axis D'. Axis D' is depicted at angle β relative to axis A. Angle

β , like angle α , represents the angle of rod **100** to plate **150** when they are assembled together. Of course, many other alternative configurations are envisioned.

(62) It is further envisioned that to accomplish the angulation of rod **100** to plate **150**, opening **182** in projection **180** may be designed at different angles as discussed above. Or as will be apparent to those skilled in the art, projection **180**, itself, may be designed on plate **150** at different angles and configurations.

(63) Optionally, plate **150** may be flat, as in FIG. 7 for example, or curved, as in FIG. 9. The curvature of plate **150** in FIG. 9 is along long axis A. However, as will be readily apparent to those skilled in the art, many other curves and shapes of plate **150** are possible, as dictated by different considerations such as manufacturability, or various clinical factors such as anatomical configurations of specific patients. Furthermore, it is contemplated that plate **150** may be provided as a rigid, non-bendable plate, or alternatively, as a deformable plate, enabling intra-operative bending, as needed.

(64) Plate **150** may be formed of any one or more suitable biocompatible materials known in the art, such as titanium, PEEK or other biocompatible materials having mechanical properties suitable for the contemplated uses of plate **150**. Plate **150** may also be coated in whole, or in part, with any suitable biocompatible material coating known in the art.

(65) For ease of reference, when rod **100** is assembled with plate **150**, this may also be referred to as a rod-plate construct, and so the rod-plate system **50** may be comprised of one or more rod-plate constructs.

(66) FIGS. 13 and 14 depict an embodiment of the rod-plate system **50**, in this case comprising multiple rod-plate constructs implanted in a human foot. Of course, it is readily recognized that the number of rod-plate constructs needed for a particular patient may be determined by the surgeon.

(67) In use, implantation of a rod-plate construct may begin by first aligning and preparing each of the targeted bones for fusion. An appropriately sized opening into the medullary canal of a targeted metatarsal is made on its dorsal aspect at its midportion. A guidewire is then passed through the opening, through the medullary canal, and continues through the other bones to be fused.

Optionally, the guidewire may be passed through to the talus or calcaneus. An appropriately-sized cannulated drill is then passed over the guidewire and used to create a passage within the bones to receive rod **100**.

(68) Rod **100**, having the selected and corresponding length to that of the drilled passage, is connected to plate **150**. The second end **106** of rod **100** is then inserted into the opening and through the passage until projection **180** of plate **150** comes to rest within the dorsal aspect opening in the metatarsal. Optionally, plate **150** may be shaped to conform to the topography of the metatarsal after insertion, or beforehand. In other embodiments, a rigid pre-contoured plate **150** may be used. Once the rod-plate construct is in place, fasteners, such as screws **164**, are then used to fixedly connect plate **150** to the metatarsal.

(69) FIGS. 15-17A depict an exemplary embodiment of shaft system **60** which is a subsystem of internal fixation system **40**. Shaft system **60** may be used independently, or in conjunction with other systems, to facilitate the alignment, reduction and fixation of bones.

(70) The prominent component of shaft system **60** is shaft **200**. Shaft **200** may be a unitary device, or a modular one comprised of various segments that may be joined together, thus enabling customization of shaft **200**. With specific reference to FIG. 15, shaft **200** is modular, and is comprised of segments referred to as cap **202**, intermediate spacer **204a**, intermediate spacer **204b**, and base **206**. These segments may each be joined to each other at a connection **203** (shown joined in FIGS. 16-17A). Connection **203** may be a threaded connection, a Morse-taper type connection, or any other suitable connection known in the art. It is further understood that one or more different types of connections **203** may be used to assemble shaft **200**.

(71) Cap **202** is a hollow body comprising a bore **215**. Cap **202** further comprises a first end **210** having a rounded terminal portion **214** which facilitates shaft **200**'s insertion into bone during

implantation, and a second end **212** where bore **215** has internal threads **216** to enable cap **202** to be connected to another meetable segment of shaft **200**. Cap **202** further has a transverse through-hole **218** to receive fixation element **220** (FIG. **16**) to fix cap **202**, and thereby facilitate fixing shaft **200**, to bone. More detail about the optional configurations and uses of through-hole **218** will be discussed in more detail later with reference to base **206**.

(72) Intermediate spacers **204a** and **204b** each have similar configurations with the difference being that spacer **204a** includes through-hole **218**, while spacer **204b** does not have any such through-holes. Both spacers **204a**, **204b** further have a first end **222**, an opposite second end **224**, and bore **215** extending therebetween, thereby making them hollow. On first end **222**, there is a projection **225** with external threads **226**. External threads **226** are configured to mate with internal threads **216**, to enable a secure connection **203** therebetween, and thus between any one spacer **204a**, **204b** and cap **202**. As is evident from this description, the intention is to enable the easy interchangeability and interconnectability of different segments.

(73) Base **206** also has a first end **232**, an opposite second end **234**, and bore **215** extending therebetween, thereby making it hollow. On its first end **232**, base **206** has a similar projection **225** with external threads **226** as discussed above with reference to other segments, for purposes of interchangeability. Base **206** is also depicted with through-hole **218** to enable its connection to bone. At its second end **234**, base **206** also has a slot **240** therethrough to facilitate bringing two bone segments together, otherwise known as reduction, as will be discussed in more detail below. Lastly, base **206** has internal threads **248** at its second end **234**. These internal threads **248** are configured to cooperate with external threads **226** of the various segments, as well as with external threads **256** of plug **208**. Notably, internal threads **248** may run deeper into bore **215** of base **206** than comparable internal threads **216** in other segments. This is because internal threads **248** are also configured to receive plug **208**.

(74) Plug **208** comprises body **255** with external threads **256** that are similar to external threads **226** on the other segments of shaft **200**, again for interchangeability, a compression element **260**, and an instrument engagement area **265**. The insertion of plug **208** into the second end **234** of base **206**, and its movement through bore **215**, together with slot **240**, and the use of a fastener **242** (FIG. **16**), enables reduction, as will be discussed yet further in more detail below.

(75) With reference to FIGS. **16-17A**, in order to perform reduction, one segment of bone must be stationary, and an adjacent segment of bone must be moved toward the stationary bone segment. In the present embodiment, bone **B1** is the stationary bone. Shaft **200** is fixed to bone **B1** with screw **220** positioned through through-hole **218** in base **206**. Notably, through-hole **218** also provides for further variability of fixation methods of shaft **200** to bone as discussed next.

(76) It is recognized that there are at least two common approaches to fixating shaft **200** in space relative to bone, among various other approaches known in the art. In one such common approach, this is accomplished by fixing proximal bone portion **Bp** to shaft **200**. In the other common approach, this is accomplished by pinning shaft **200** in between two segments of bone.

(77) With reference to the first approach, when shaft **200** is in a bone canal (FIG. **17**), a bore **230** is drilled from the proximal bone surface **Bp** of proximal bone **B1** to the exterior of shaft **200**. Bore **230** is coaxial with through-hole **218** and sized to allow the head of screw **220** to pass. Screw **220** is then inserted through bore **230** and threaded into a threaded version of through-hole **218**. As is evident, and known in the art, it is not necessary for screw **220** to go all the way through base **206**, so long as there is sufficient threaded engagement in the first area of contact between screw **220** and base **206**. Of course, if more thread engagement is desired, screw **220** may be progressed further to threadably engage base **206** at its second area of contact with base **206**. As will be readily recognized, the approach just described from bone surface **Bp** may be similarly accomplished from the opposite distal bone surface **Bd**. For this reason, it is desirable to have through-hole **218** fully threaded.

(78) With respect to the second approach of fixating shaft **200** in space relative to bone, the

intention is for screw **220** to enter through bore **230**, completely extend through shaft **200**, and threadably purchase bone portion **Bd**. While this approach may better call for through-hole **218** to be unthreaded, to increase the variability of usage of shaft **200**, the threaded version of through-hole **218** may continue to be employed. Of course, having described the foregoing, it will be readily recognized that various alternatives and permutations of the above configurations of structure and usage may be employed. Additionally, it is noted that through-hole **218** has the same characteristics as through-hole **130** of rod **100**.

(79) With further reference to FIGS. **16-17A**, in the context of reduction, the bone segment that will be moved toward stationary bone segment **B1** is movable bone segment **B2**. Of course, it is generally recognized that movement is relative. In similar fashion as described for putting screw **220** through through-hole **218** of base **206**, a partially threaded screw **242** is put through slot **240**, and specifically through slot area **244** which is most distal to base **206**'s first end **232**, and screwed into bone **B2**. Plug **208** may be introduced into threaded engagement with internal threads **248** of second end **234** of base **206** before or after insertion of screw positioned at slot area **244**. A driver (not shown) is then engaged to plug **208**'s instrument engagement area **265**, and actuated so as to rotate plug **208** in direction **F** (FIG. **17A**), thus translating plug **208** towards slot area **246** by virtue of the translational component of threaded rotation. In so doing, compression element **260** of plug **208** pushes screw **242** linearly along slot **240** from slot area **244** to slot area **246**, a distance **M**. Consequently, this pushes bone segment **B2** towards bone segment **B1**, which is fixedly connected to base **206** via through-hole **218**, the same distance **M**, thus producing reduction.

(80) It will be readily understood by those skilled in the art that the reduction mechanism described above, inclusive of such elements as slot **240** and plug **208**, may be positioned on any other one or more segments of shaft **200** for increased variability of the segments of shaft **200**. Of course, other means to rotate or otherwise advance plug **208** may be needed in instances where its instrument engagement area **265** is not accessible as in the embodiment described above.

(81) While slot **240** has been described in the context and functionality depicted with reference to shaft **200**, a somewhat similar slot **132** is positioned on rod **100** (FIG. **5**) as mentioned earlier. The specific mechanism for reduction using slot **132** will be different from that described for slot **240**, but such other ways to perform reduction are known in the art. For example, with reference to FIG. **24**, if a pin **310** were inserted through slot **132** of rod **100** into one bone, and a screw **220** were inserted through hole **130** of rod **100** into another bone, then a surgeon may manually grip and translate pin **310** and screw **220** toward each other, thereby bringing the two bones together.

(82) The ability to rotationally align one segment of shaft **200** to a desired position relative to another segment of shaft **200**, for a variety of purposes, is an engineering function that has many different solutions known in the art. For example, with reference to FIG. **16**, it may be desirable to have through-hole **218** on cap **202** oriented 30° in either direction from where it is shown.

Similarly, on the same shaft **200**, it may be desired for through-hole **218** on spacer **204a** to be oriented 15° from where it is shown. One way to accomplish such desired alignment of through-holes **218** in an assembled shaft **200** is to have keyed, timed, or precision threads **216**, **226** that enable planned rotational assembly to result in the desired alignment.

(83) The ability to target and insert screws **220**, **242** into their respective positions in or through rods **100** and shafts **200** is also an engineering function that has various different solutions known in the art. For example, targeting jigs are known to facilitate the location and identify the orientation of through-holes **218** when shaft **200** is inside bone. Rods **100** and shafts **200** may be configured to cooperate with such targeting jigs.

(84) As was noted earlier with respect to rod **100**, shaft **200**, and the various other elements described above, may be formed of any suitable material known in the art. For example, shaft **200** may be formed of titanium, or other biocompatible materials having mechanical properties suitable for its contemplated uses. Furthermore, shaft **200** may be coated with any suitable biocompatible coating known in the art, such as hydroxyapatite or the like, or may be uncoated, as needed to suit

particular mechanical and clinical needs.

(85) Notably, as will be apparent to those skilled in the art, shafts **200** may be solid rather than hollow, and rods **100** may be hollow rather than solid. Of course, various other adjustments to their respective features may then be made to result in those features maintaining their respective intended functionalities. For example, if shaft **200** were solid rather than hollow, second end **234** of base **206** would still maintain a hollow passageway to enable plug **208** to travel therethrough to effectuate reduction.

(86) Referring to FIGS. **18** and **19**, two shafts **200** of shaft system **60** are shown implanted in a human foot. The medial shaft **200** is assembled from a cap **202**, followed by two spacers **204a**, and then a base **206**, and is oriented such that base **206** is located in the first metatarsal, and the cap **202** is in the talus. The lateral shaft **200** is assembled from another cap **202** and a base **206**. It is oriented such that base **206** is in the calcaneus, and cap **202** is in the cuboid. Each shaft **200** is depicted as fixed in place to bone by various screws **220**, **242**.

(87) Each of the bones targeted for fusion are first manually aligned and prepared for accepting shaft system **60**. For example, to implant the medial shaft **200**, the first metatarsal phalangeal joint is exposed via a dorsal incision. A guidewire to direct reaming of the medullary canal is then introduced near the center of the metatarsal head, directed through the metatarsal body, medial cuneiform navicular, and into the talar neck. Reaming is then conducted iteratively over the guidewire until a canal having an appropriate internal diameter to receive shaft **200** is formed through the bones. Of course, it is recognized that various anatomical landmarks and sizes of bones will ultimately determine the selection of the diameter and length of the canal, and therefore the diameter and length of shaft **200** to be used in it, as well as which segments of shaft **200** should be selected and the order in which they should be assembled.

(88) Prior to insertion of shaft **200**, it may be connected to a targeting jig (not shown). The targeting jig projects the positions of the relevant through-holes on shaft **200**, such as holes **218**, and thus enables a surgeon to accurately place screws **220** through the targeting jig directly into or through each targeted hole **218**, for example, as the case may be, after shaft **200** is placed in the reamed bone canal.

(89) Once the targeting jig is attached to the medial shaft **200**, medial shaft **200** is then inserted into the reamed bone canal. A screw **220** is then placed through the jig and through hole **218** of cap **202** which is located in the talus. Then, screw **242** is placed through the jig and through slot **240**, which, in such embodiment, is in the first metatarsal. Plug **208**, located in second end **234** of base **206** is then rotationally actuated to translate axially along slot **240**, thus compressing all the bones between screw **242** and screw **220** along shaft **200**, to a desired orientation, at which point, other screws **220** will be inserted through shaft **200** to lock the compressed bones in place relative to shaft **200**. Alternatively, some or all of the compression may also be performed by other techniques known in the art.

(90) In light of the foregoing, the preparation of bones, the selection, assembly, and insertion of lateral shaft **200**, and the reduction and fixation of the associated bones, will be apparent to those skilled in the art.

(91) With reference to FIGS. **20-21**, there is depicted the implanted combination of a rod-plate system **50** and shaft system **60**. Specifically, there are two rod-plate constructs, and two shafts **200**. Notably, a rod **100** may be connected to a shaft **200**, and is depicted as such. This may be accomplished using screw **220**, or by other means. As will be apparent to those skilled in the art, any number, combination, and configuration of rod-plate constructs and shafts may be employed, and also connected to each other at different points, as necessary to address various clinical needs.

(92) FIG. **22** depicts a midfoot component in the form of plate **280**, having screw holes **281** adapted to receive screws **282** (FIG. **23**) that attach plate **280** to bone. Plate **280** is generally known in the art. Plate **280** may be malleable, to enable it to be shaped in situ to conform to target anatomy to which it will be attached, or provided in multiple generic rigid shapes, or otherwise custom

manufactured through patient-matching technologies. It is intended to be used on its own, or in combination with one or more systems and subsystems of fixation system **10**.

(93) FIG. **23** depicts plate **280** implanted in a foot in combination with rod-plate constructs and shafts **200**, wherein certain of the rods **100** and shafts **200** are interconnected. The possibility to combine all these subsystems offers increased variability and flexibility to treat a greater variety of clinical needs.

(94) FIG. **24** depicts an internal fixation subsystem, namely shaft system **60**, connected to an external fixation system **80**. External fixation system **80** comprises one or more frames **302** each having a plurality of openings **304**, a plurality of connectors **306**, and pins **310**. Pins **310** are depicted as partially threaded on their distal ends, and may also optionally be unthreaded, or fully threaded, as is known in the art. This basic external fixation system **80** is also well known in the art.

(95) Notably, an aspect of the present invention is the connection of external fixation system **80** to the shaft system **60**, as depicted in FIG. **24**. Specifically, frame **302** is connected using pins **310** inserted through connectors **306** to shafts **200**. It is thusly evident that many other configurations, arrangements and connections of external fixation system **80** with one or more of certain subsystems of internal fixation system **40** are possible. For example, external fixation system **80** may be connected to an implanted rod-plate system **50**.

(96) The added ability to combine and connect external fixation system **80** with the various subsystems of internal fixation system **40** increases yet further the variability and flexibility of the overall fixation system **10** to treat a yet greater variety of clinical needs.

(97) FIG. **25** depicts a sole component **400** of external fixation system **80**. Sole component **400** is intended to be used in conjunction with external fixation system **80** and is meant to provide a weight-bearing platform underneath a patient's foot to enable a patient with fixation system **10** to walk.

(98) Sole **400** comprises a housing **402** adapted to hold a removable liner **404** therein. With reference from front **410** to back **412**, housing **402** has a top surface **406**, side walls **407** and a bottom surface **408** that spans housing **402** from front **410** to back **412**. Bottom surface **408** is the surface that comes in contact with the ground when a patient walks while wearing fixation system **10**. As such, its shape, texture and materials may be adapted as known in the art to facilitate safer walking. For example, bottom surface **408** may be made of, or coated with, rubber to increase the coefficient of friction between sole **400** and the ground, thus diminishing the chances of a patient slipping while they walk with fixation system **10**. Other embodiments are envisioned to accomplish this goal.

(99) The profile of bottom surface **408** may be a complex series of continuous curves, such as a tighter curve toward the back **412**, which may be known as the heel-strike area, eventually transitioning to a gradual curve toward the front **410**, over which the forefoot rolls during gait. Of course, other shapes are contemplated.

(100) Liner **404** may include an inflatable air bladder (not specifically shown) and be filled with any appropriate fluid. The bladder may have a valve (not specifically shown) through which air or other fluid may be introduced or evacuated to achieve the optimal density and size to support a particular weight or pressure requirement. Liner **404** may optionally be housed in a fabric shell (not specifically shown). The shell can be moisture wicking and machine washable for easy cleaning and maintenance. The shell may also be removable via a zipper, hook-and-loop fasteners, snaps, and the like.

(101) Sole **400** also has slots **424**, enabling it to be connected to frame **302** (FIG. **26**), as will be discussed in more detail below. Slots **424** may be formed within the material of housing **402**, or may alternatively be solid tubes affixed to either the inside or outside of housing **402**, as may be readily understood by those skilled in the art.

(102) FIG. **26** is an exploded view of certain components of external fixation system **80** used to assemble sole **400** to frame **302**. More specifically, connectors **306** have threaded openings **307** on

their various faces, enabling them to be connected to frame 302 and to other components using bolts 308, thus making them universal.

(103) Struts 426 are used to connect sole 400 to frame 302. Struts 426 have eyelets 428 to enable such a connection. For example, after connectors 306 are connected to frame 302 with bolts 308, eyelets 428 of struts 426 would be aligned with connectors 306, and additional bolts 308 would be put through eyelets 428 and screwed into connectors 306, thus pinning and fixing struts 426 to frame 302. Struts 426 shall also be connected to sole 400 in any variety of ways known to those skilled in the art. For example, struts 426 may be inserted and glued into slots 424.

(104) Struts 426 may be of unitary construction, or alternatively, may be comprised of two or more components enabling strut 426 to expand and compress as well as be fixed in place at a desired length. Struts 426 may also be made from various materials, from stiff metals, to more elastomeric materials that may further facilitate absorption of striking forces during gait.

(105) Having thus described the invention in detail, it is to be understood that the foregoing description is not intended to limit the spirit or scope thereof. It will be understood that the embodiments of the present disclosure described herein are merely exemplary and that a person skilled in the art may make any variations and modification without departing from the spirit and scope of the disclosure. All such variations and modifications, including those discussed above, are intended to be included within the scope of the disclosure.

(106) Although the invention has been described by reference to its preferred embodiment as is disclosed in the specification and drawings above, many more embodiments of the invention are possible without departing from the invention. Thus, the scope of the invention should be limited only by the appended claims.

Claims

1. A fixation system for immobilizing a skeletal structure, comprising: an internal fixation system having a rod-plate system, and a shaft system; and an external fixation system; wherein the rod-plate system comprises a rod affixed to a plate without a separate connecting member therebetween, the rod being adapted to be positioned in a bone canal, and the plate being adapted to be positioned on bone near the bone canal, the plate further having a first side adapted to face bone, an opposite second side, a length, a width, a plate axis along the length, and a projection extending from the first side, the projection having a cylindrical opening with a longitudinal opening axis, and the opening being oriented such that the opening axis is at an angle to the plate axis; wherein the rod-plate system is connected to the shaft system with a fixation element when both systems are located in bone; and wherein the external fixation system is connected to one of the rod-plate system or shaft system.
2. The fixation system of claim 1, wherein the rod of the rod-plate system is modularly comprised of multiple segments each joinable by a connection.
3. The fixation system of claim 2, wherein the connection is either a threaded connection or a Morse-taper connection.
4. The fixation system of claim 1, wherein the external fixation system further comprises a pin, and wherein the pin connects the external fixation system to one of the rod-plate system or shaft system.
5. The fixation system of claim 4, wherein the external fixation system further comprises a frame connected to a sole, the sole having a bottom adapted to contact ground.
6. The fixation system of claim 1, further comprising a midfoot plate system attached to bone, the midfoot plate system comprising a plate and a fastener.
7. The fixation system of claim 1, wherein the shaft system comprises a shaft with a longitudinal axis, a slot on the shaft oriented in the direction of the longitudinal axis, and a hole on the shaft oriented at an angle to the longitudinal axis, the shaft further adapted to be positioned in a bone

- canal and configured to move two bone segments that comprise the bone canal toward each other.
8. The fixation system of claim 7, wherein the fixation element connecting the rod to the shaft is an at least partially threaded fastener.
9. The fixation system of claim 7, wherein the rod is implanted in one metatarsal of a foot and the shaft is implanted in a different metatarsal of the same foot.
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